Homework #3 — Trading and Portfolio Optimization

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Statistical Signal Processing and Learning, AA2025-2026

A portfolio of 10 NASDAQ technology stocks is traded over 2018. The first semester (Jan-Jun, ~124 trading days) trains the models; the second semester (Jul-Dec) evaluates them out of sample. Starting capital: 10.000 EUR.

Data loading

```
clc; clear; close all;
load("DataStocks/Hw3.mat");

open_prices   = table();
close_prices  = table();
total_volumes = table();

for i = 1:length(Stock)
    ticker = strtrim(Stock(i,:));
    td = eval(ticker);
    open_prices.(ticker)   = td(2:end,1);
    close_prices.(ticker)  = td(2:end,4);
    total_volumes.(ticker) = td(2:end,5);
end

T_days   = height(open_prices);
n_stocks = width(open_prices);
tickers  = open_prices.Properties.VariableNames;
P_o  = table2array(open_prices);
P_c  = table2array(close_prices);
Vol  = table2array(total_volumes);

N_train = 124;
idx_tr  = 1:N_train;
idx_te  = (N_train+1):T_days;
C0 = 10000;
```

1 — Linear Price Predictor

For each stock k, the closing price is estimated at the opening of day t:

$$\hat{P}_k(t+1|t) = \mathbf{b}_k^T \mathbf{x}_k(t)$$

The feature vector uses the current opening price, the four most recent closing prices, and the previous day's volume:

$$\mathbf{x}_k(t) = \left[1,\, P_k^{(o)}(t),\, P_k^{(c)}(t-1),\, \ldots,\, P_k^{(c)}(t-4),\, V_k(t-1)/10^6\right]^{T}$$

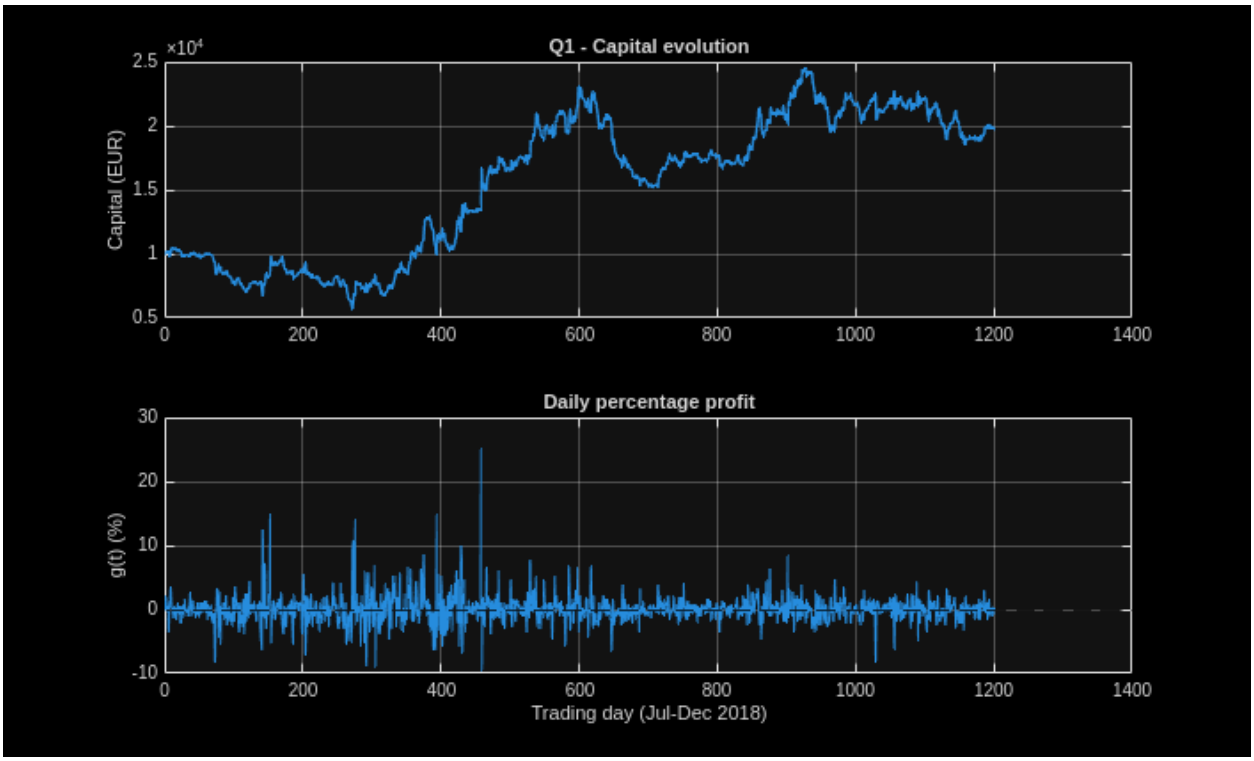
Coefficients b_k are estimated by ordinary least squares on the training set. At each test day the entire capital is placed on the single stock with the highest predicted intraday gain $(\hat{P}_k - P_k^{(o)})/P_k^{(o)}$.

```
L = 5;
B = zeros(L+2, n_stocks);

for k = 1:n_stocks
    valid = idx_tr(idx_tr > L);
    Xtr   = zeros(length(valid), L+2);
    ytr   = zeros(length(valid), 1);
    for ii = 1:length(valid)
        t = valid(ii);
        Xtr(ii,:) = [1, P_o(t,k), P_c(t-1,k), P_c(t-2,k), ...
                        P_c(t-3,k), P_c(t-4,k), Vol(t-1,k)/1e6];
        ytr(ii) = P_c(t,k);
    end
    B(:,k) = (Xtr'*Xtr) \ (Xtr'*ytr);
end

% Trading simulation
C  = zeros(length(idx_te)+1, 1);  C(1) = C0;
pred_g = zeros(length(idx_te), n_stocks);
act_g  = zeros(length(idx_te), n_stocks);

for ii = 1:length(idx_te)
    t = idx_te(ii);
    P_hat = zeros(n_stocks,1);
    for k = 1:n_stocks
        x_t = [1, P_o(t,k), P_c(t-1,k), P_c(t-2,k), ...
                  P_c(t-3,k), P_c(t-4,k), Vol(t-1,k)/1e6]';
        P_hat(k) = B(:,k)' * x_t;
    end
    pred_g(ii,:) = (P_hat' - P_o(t,:)) ./ P_o(t,:);
    act_g(ii,:)  = (P_c(t,:) - P_o(t,:)) ./ P_o(t,:);
    [bg, bk] = max(pred_g(ii,:));
    if bg > 0
        C(ii+1) = C(ii) * P_c(t,bk) / P_o(t,bk);
    else
        C(ii+1) = C(ii);
    end
end
g_t = diff(C) ./ C(1:end-1);

fprintf('Q1 - Final capital: %.2f EUR  (+%.1f%%)\n', C(end), 100*(C(end)-C0)/
C0);
```

```
Q1 - Final capital: 19763.31 EUR  (+97.6%)
```

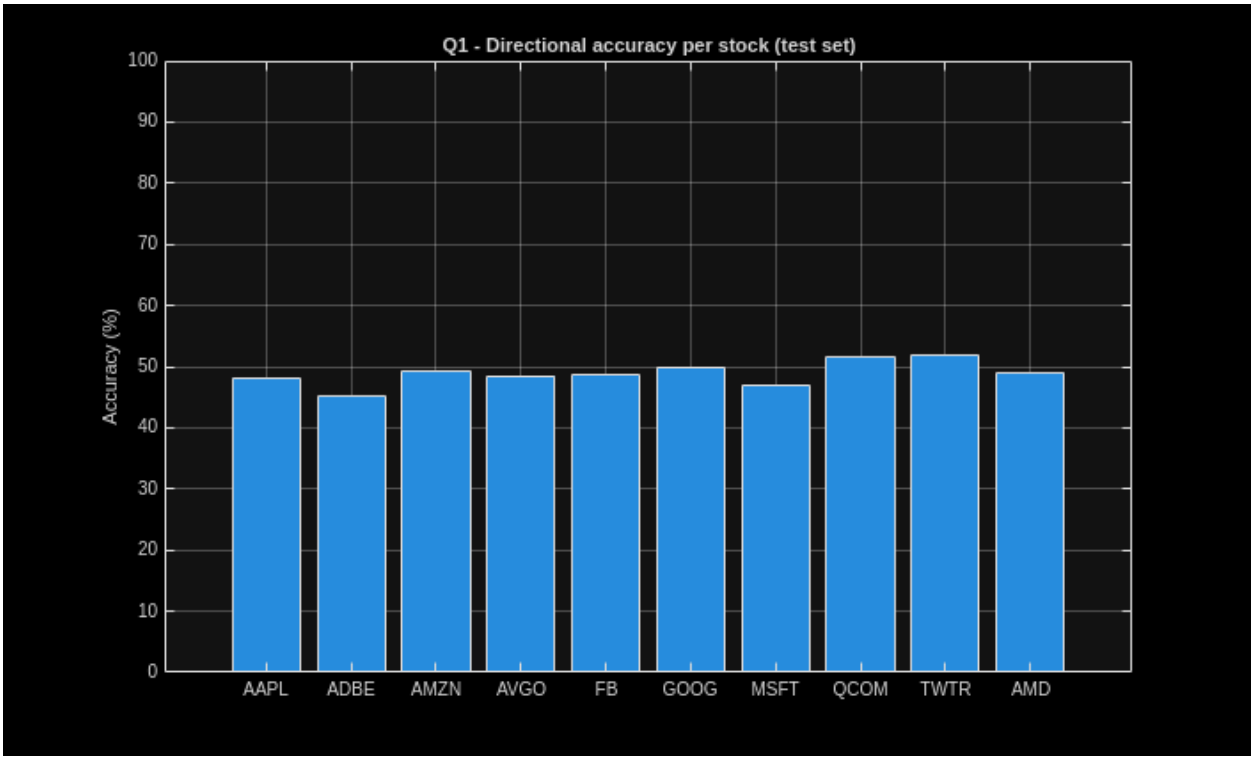
Capital evolution $C(t)$ and daily percentage profit $g(t) = \Delta C(t)/C(t)$.

```
figure;
subplot(2,1,1);
plot(1:length(C), C, 'LineWidth', 1.5);
ylabel('Capital (EUR)'); title('Q1 - Capital evolution'); grid on;

subplot(2,1,2);
plot(1:length(g_t), g_t*100, 'LineWidth', 1.2); yline(0,'k--');
xlabel('Trading day (Jul-Dec 2018)');
ylabel('g(t) (%)'); title('Daily percentage profit'); grid on;
```



Directional prediction accuracy: fraction of test days where the sign of the predicted gain matches the actual intraday move.

```
correct1 = zeros(1, n_stocks);
for ii = 1:length(idx_te)
    correct1 = correct1 + ((act_g(ii,:)>0) == (pred_g(ii,:)>0));
end

figure;
bar(correct1 / length(idx_te) * 100);
xticks(1:n_stocks); xticklabels(tickers);
ylabel('Accuracy (%)');
title('Q1 - Directional accuracy per stock (test set)');
ylim([0 100]); grid on;
```



2a — Logistic Regression

The binary label $y_i(t) \in \{0, 1\}$ marks up-days: $y_i(t) = 1$ iff $P_i^c(t) > P_i^o(t)$. Buy probability is modelled by:

$$h_{\mathbf{b}_i}(\mathbf{x}_i(t)) = \frac{1}{1 + e^{-\mathbf{b}_i^\top \mathbf{x}_i(t)}}$$

with the same feature vector as Q1. Parameters $\mathbf{b}_i$ are found by gradient descent on the binary cross-entropy (2000 steps, step size $\alpha = 0.1/N$). All features are standardised before training.

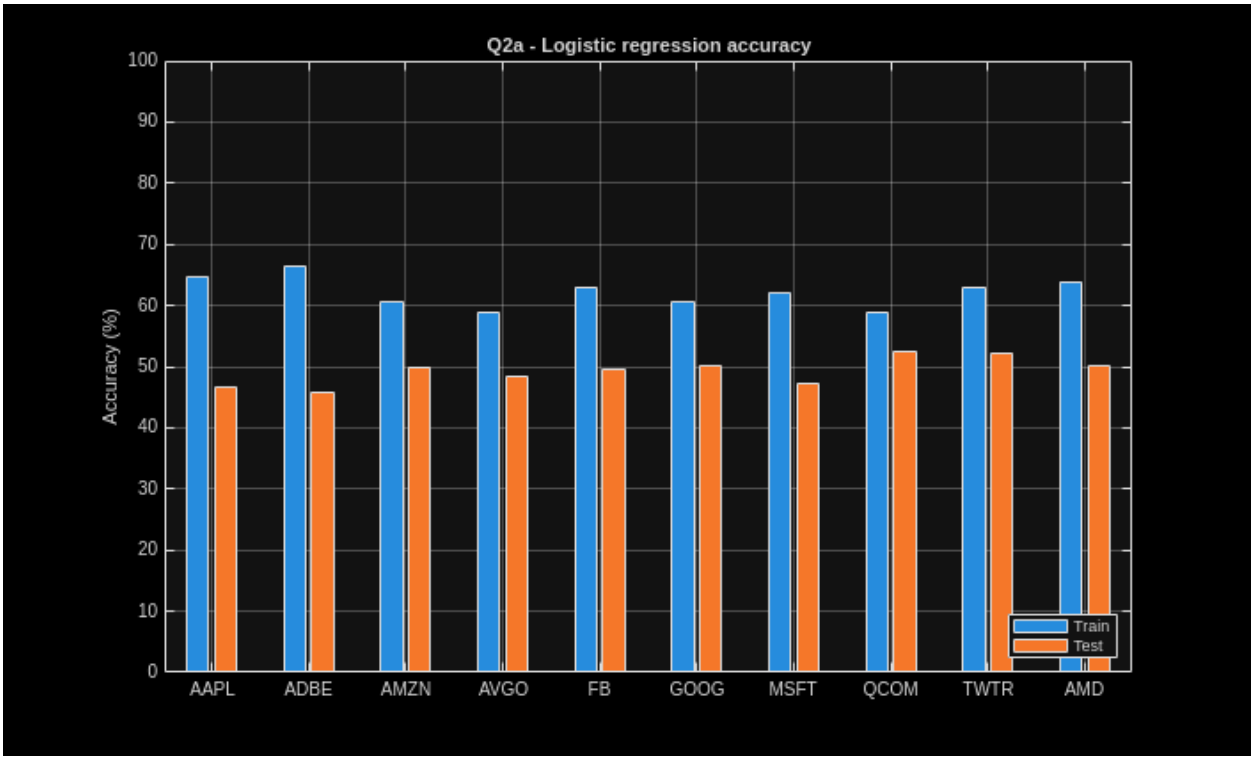
```
sigmoid  = @(z) 1 ./ (1+exp(-z));
B_log    = zeros(L+2, n_stocks);
mu_all   = zeros(n_stocks, L+1);
sig_all  = zeros(n_stocks, L+1);
acc_tr   = zeros(1, n_stocks);
acc_te   = zeros(1, n_stocks);
h_tr_all = cell(n_stocks,1);
y_tr_all = cell(n_stocks,1);

for k = 1:n_stocks
    valid_tr = idx_tr(idx_tr > L);
    N_tr = length(valid_tr);
    Xtr = zeros(N_tr, L+2);  ytr = zeros(N_tr,1);
    for ii = 1:N_tr
        t = valid_tr(ii);
        Xtr(ii,:) = [1, P_o(t,k), P_c(t-1,k), P_c(t-2,k), ...
                        P_c(t-3,k), P_c(t-4,k), Vol(t-1,k)/1e6];
        ytr(ii) = P_c(t,k) > P_o(t,k);
    end
```

```
    mu_x  = mean(Xtr(:,2:end));
    sig_x = std(Xtr(:,2:end));
    Xtr_n = [ones(N_tr,1), (Xtr(:,2:end)-mu_x)./sig_x];
    b = zeros(L+2,1);
    for it = 1:2000
        hh = sigmoid(Xtr_n*b);
        b  = b - (0.1/N_tr) * (Xtr_n'*(hh-ytr));
    end
    B_log(:,k)   = b;
    mu_all(k,:)  = mu_x;
    sig_all(k,:) = sig_x;
    h_tr_all{k}  = sigmoid(Xtr_n*b);
    y_tr_all{k}  = ytr;
    acc_tr(k) = mean((h_tr_all{k} > 0.5) == ytr) * 100;

    valid_te = idx_te(idx_te > L);  N_te = length(valid_te);
    Xte = zeros(N_te,L+2);  yte = zeros(N_te,1);
    for ii = 1:N_te
        t = valid_te(ii);
        Xte(ii,:) = [1, P_o(t,k), P_c(t-1,k), P_c(t-2,k), ...
                        P_c(t-3,k), P_c(t-4,k), Vol(t-1,k)/1e6];
        yte(ii) = P_c(t,k) > P_o(t,k);
    end
    Xte_n = [ones(N_te,1), (Xte(:,2:end)-mu_x)./sig_x];
    acc_te(k) = mean((sigmoid(Xte_n*b) > 0.5) == yte) * 100;
end

fprintf('\n%-6s  Train   Test\n','Stock');
for k = 1:n_stocks
    fprintf('%-6s  %5.1f%%  %5.1f%%\n', tickers{k}, acc_tr(k), acc_te(k));
end
```

```
Stock   Train   Test
AAPL     64.7%   46.5%
ADBE     66.4%   45.8%
AMZN     60.5%   50.0%
AVGO     58.8%   48.3%
FB       63.0%   49.5%
GOOG     60.5%   50.2%
MSFT     62.2%   47.2%
QCOM     58.8%   52.4%
TWTR     63.0%   52.2%
AMD      63.9%   50.1%
```

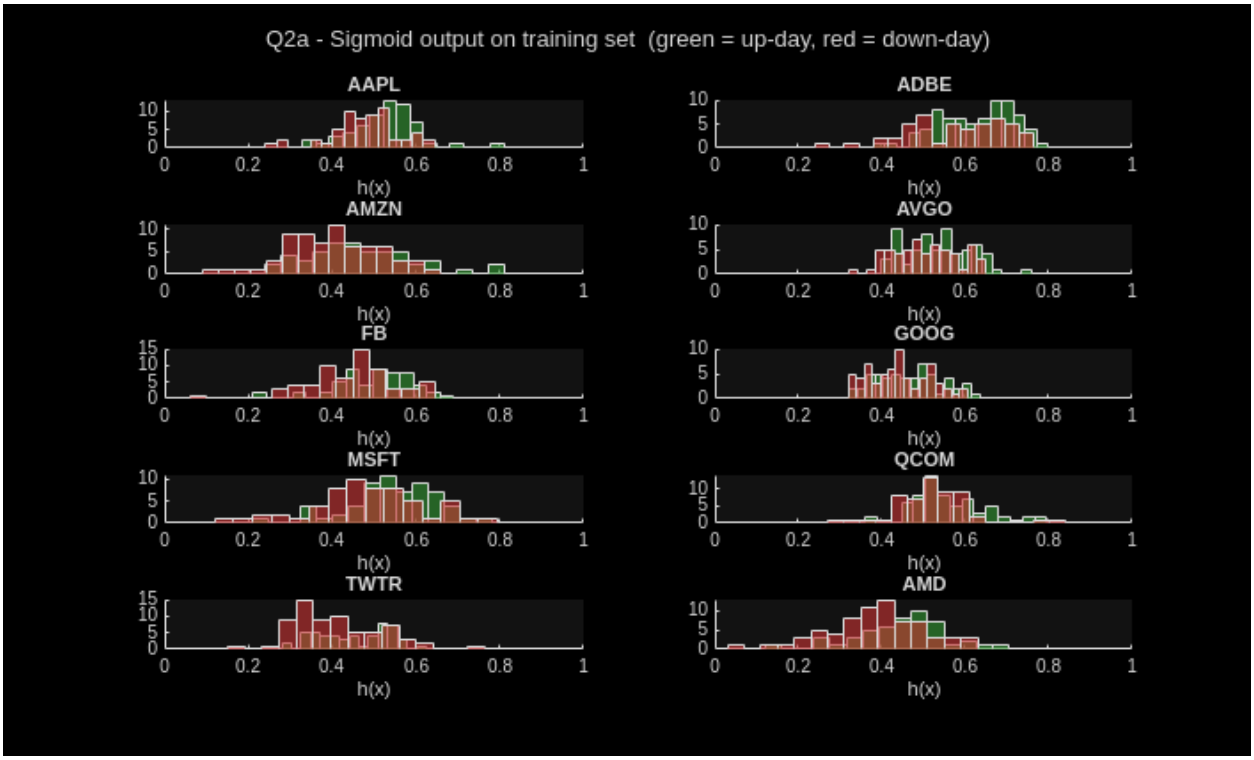
Training accuracy reaches 60-68%; test accuracy drops to near 50%. The model captures in-sample patterns but stock direction is essentially unpredictable out of sample, especially in the volatile second half of 2018.

```
figure;
bar([acc_tr; acc_te]');
xticks(1:n_stocks); xticklabels(tickers);
ylabel('Accuracy (%)'); title('Q2a - Logistic regression accuracy');
legend({'Train','Test'},'Location','southeast'); ylim([0 100]); grid on;
```



Sigmoid output distributions on the training set. A well-calibrated model would shift the green (up-day) mass towards 1 and the red (down-day) mass towards 0. The significant overlap confirms limited separability.

```
figure;
for k = 1:n_stocks
    subplot(5,2,k); hold on;
    histogram(h_tr_all{k}(y_tr_all{k}==1), 15, 'FaceColor',[0.2 0.6 0.2]);
    histogram(h_tr_all{k}(y_tr_all{k}==0), 15, 'FaceColor',[0.8 0.2 0.2]);
    title(tickers{k}); xlabel('h(x)'); xlim([0 1]); hold off;
end
sgtitle('Q2a - Sigmoid output on training set  (green = up-day, red = down-
day)');
```



2b — Portfolio Strategies

Three allocation rules are tested on the test period using the buy probabilities $h_k(t)$ produced by the logistic models:

- **S1 - winner-takes-all:** all capital on $\arg\max_k h_k(t)$ provided $\max_k h_k > 0.5$; otherwise hold.
- **S2 - proportional:** split capital proportional to h_k among all stocks with $h_k > 0.5$; hold if none qualify.
- **S3 - high-confidence:** same as S2 with the threshold raised to $h_k > 0.6$, keeping only the most confident predictions.

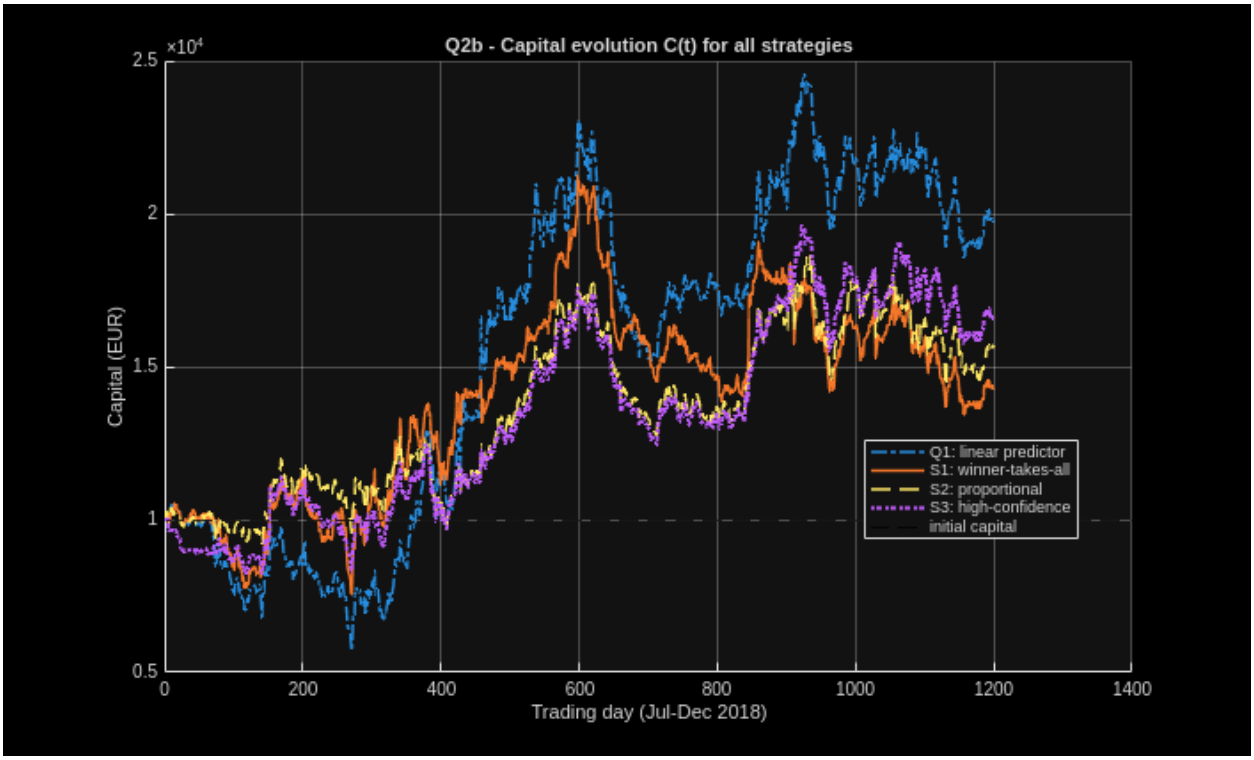
```
N_te_s = length(idx_te);
C1 = zeros(N_te_s+1,1);  C1(1) = C0;
C2 = zeros(N_te_s+1,1);  C2(1) = C0;
C3 = zeros(N_te_s+1,1);  C3(1) = C0;

for ii = 1:N_te_s
    t = idx_te(ii);
    h = zeros(n_stocks,1);
    for k = 1:n_stocks
        x_t = [1, (P_o(t,k)      -mu_all(k,1))/sig_all(k,1), ...
                  (P_c(t-1,k)    -mu_all(k,2))/sig_all(k,2), ...
                  (P_c(t-2,k)    -mu_all(k,3))/sig_all(k,3), ...
                  (P_c(t-3,k)    -mu_all(k,4))/sig_all(k,4), ...
                  (P_c(t-4,k)    -mu_all(k,5))/sig_all(k,5), ...
                  (Vol(t-1,k)/1e6-mu_all(k,6))/sig_all(k,6)]';
        h(k) = sigmoid(B_log(:,k)' * x_t);
    end
    ret = P_c(t,:)' ./ P_o(t,:)';
```

```
    [bh, bk] = max(h);
    if bh > 0.5; C1(ii+1)=C1(ii)*ret(bk); else; C1(ii+1)=C1(ii); end

    m2 = h > 0.5;
    if any(m2); w=h.*m2/sum(h.*m2); C2(ii+1)=C2(ii)*(w'*ret); else;
C2(ii+1)=C2(ii); end

    m3 = h > 0.6;
    if any(m3); w=h.*m3/sum(h.*m3); C3(ii+1)=C3(ii)*(w'*ret); else;
C3(ii+1)=C3(ii); end
end

fprintf('\nQ2b - Final capitals:\n');
fprintf('  S1 winner-takes-all  %.2f EUR  (%+.1f%%)\n', C1(end), 100*(C1(end)-
C0)/C0);
fprintf('  S2 proportional      %.2f EUR  (%+.1f%%)\n', C2(end), 100*(C2(end)-
C0)/C0);
fprintf('  S3 high-confidence   %.2f EUR  (%+.1f%%)\n', C3(end), 100*(C3(end)-
C0)/C0);
```

```
Q2b - Final capitals:
  S1 winner-takes-all  14289.03 EUR  (+42.9%)
  S2 proportional      15678.32 EUR  (+56.8%)
  S3 high-confidence   16616.66 EUR  (+66.2%)
```

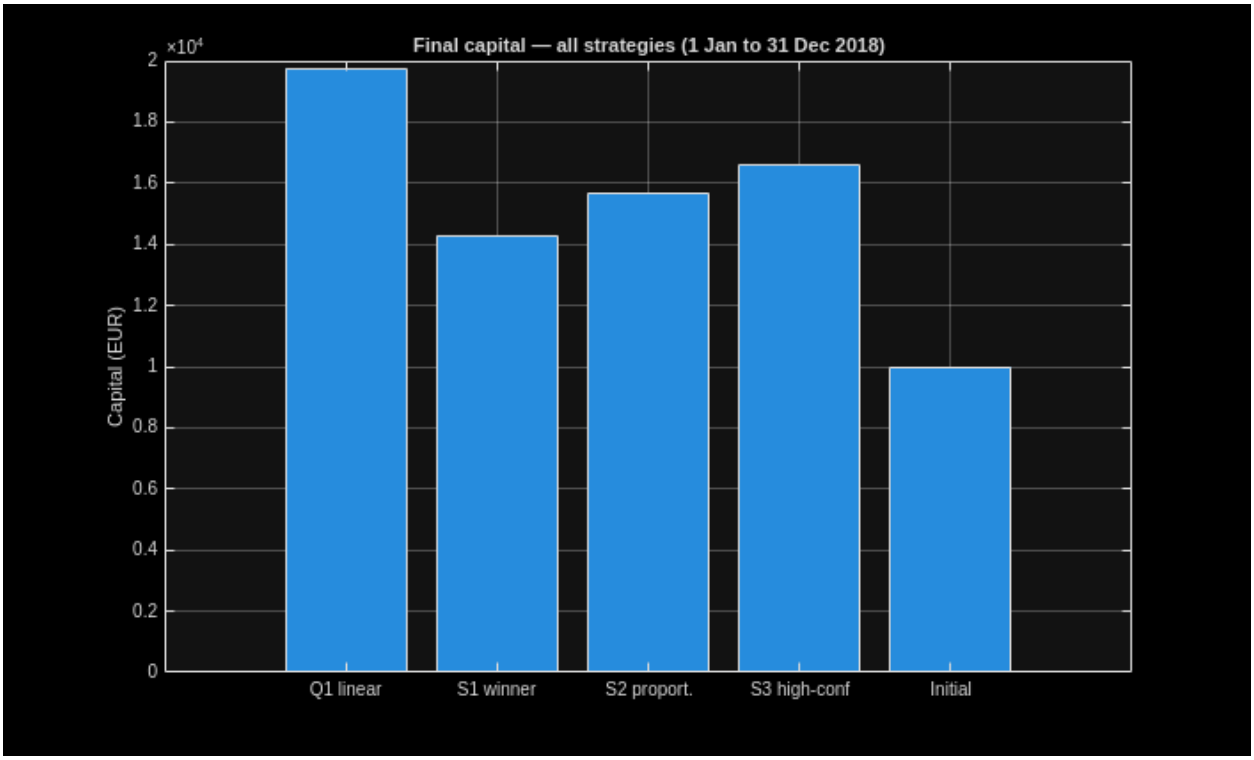
Capital evolution $C(t)$ from Jul-Dec 2018 for all four strategies. The Q1 linear predictor outperforms all logistic strategies due to its aggressive single-stock allocation and good performance on a few large up-moves early in the test period.

```
t_ax = 0:N_te_s;
figure;
hold on;
plot(t_ax, C,  '-.',  'LineWidth',1.5, 'DisplayName','Q1: linear predictor');
plot(t_ax, C1, '-',   'LineWidth',1.5, 'DisplayName','S1: winner-takes-all');
plot(t_ax, C2, '--',  'LineWidth',1.5, 'DisplayName','S2: proportional');
plot(t_ax, C3, ':',   'LineWidth',2,   'DisplayName','S3: high-confidence');
yline(C0,'k--','LineWidth',1,'DisplayName','initial capital');
xlabel('Trading day (Jul-Dec 2018)'); ylabel('Capital (EUR)');
title('Q2b - Capital evolution C(t) for all strategies');
legend('Location','best'); grid on; hold off;
```



Summary of final capitals. Strategies differ substantially in risk profile: S1 concentrates risk (like Q1) while S2/S3 spread it across multiple stocks, trading peak gains for lower volatility.

```
figure;
bar([C(end), C1(end), C2(end), C3(end), C0]);
xticklabels({'Q1 linear','S1 winner','S2 proport.','S3 high-conf','Initial'});
ylabel('Capital (EUR)');
title('Final capital — all strategies (1 Jan to 31 Dec 2018)');
grid on;
```



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